

NANDINI BHUVA

☎ 857-492-5872 ✉ bhuva.n@northeastern.edu  [linkedin.com/in/nandinibhuva](https://www.linkedin.com/in/nandinibhuva)

Education

Northeastern University – Boston, MA, USA

Sep 2024 – Dec 2026

Master of Science in Cybersecurity

GPA – 3.67

Relevant Courses: Network Security, Software Vulnerability and Security, Network security, Cybersecurity principles, Information system Forensics

Pandit Deendayal Petroleum University – Gandhinagar, INDIA

Aug 2019 – May 2023

Bachelor of Technology in Information and Communication Technology

GPA – 4.00

Relevant Courses: Information Security, Computer Communication and Networks, Cloud Architecture and Services

Work Experience

Teaching Assistant

Sep 2025 – Present

Northeastern University

Boston, MA

- Support 70+ students in coursework through weekly office hours and labs, collaborating with teaching staff to develop educational materials and communicating complex technical concepts to diverse learners.

Information Security Analyst

Jun 2023 – Jun 2024

DI Solutions

Surat, India

- Monitored and investigated 5–10 weekly security alerts across firewalls, cloud workloads, and 40+ endpoints using log analysis; documented findings in clear, non-technical language for stakeholders and followed incident response protocols.
- Conducted monthly vulnerability assessments on 20+ internal and client systems using automated scanners; validated findings manually, ranked risks using CVSS, and coordinated patch cycles to reduce exploitable attack surface.
- Strengthened client security posture by implementing MFA, enforcing password & access policies, and completing technical security questionnaires for audits, compliance reviews, and incident documentation.
- Managed endpoint protections including antivirus tools, encryption verification, lock policies, and automatic updates across 40+ devices; maintained organized audit records documenting configurations, changes, and security outcomes.

Full Stack Development Engineer

May 2020 – Jul 2020

TalentServe

India

- Developed a secure login timer system applying secure software development principles including input validation, session management, and access control mechanisms to ensure data integrity and prevent unauthorized access.

Projects

Web Application Security Testing

- Conducted penetration testing on web applications using Burp Suite and OWASP ZAP, exploiting SQL injection vulnerabilities to extract database credentials and XSS flaws to achieve session hijacking and privilege escalation.
- Developed custom Python exploit scripts to automate SQLi enumeration and XSS payload delivery; reverse-engineered authentication mechanisms to bypass broken access controls and identify logic flaws in session management.

Network Segmentation and Traffic Monitoring

- Architected a multi-tier network using Linux routers, VLANs, iptables, and NAT rules to enforce segmentation between user, application, and database zones, reducing lateral movement exposure.
- Performed active traffic analysis, detecting malformed packets, scan attempts, and misconfigured services; validated rule robustness with SYN flood tests, port-scan evasion techniques, and ICMP rate-limiting.

Malware Analysis & Memory Forensics Investigation

- Performed using IDA Pro, Ghidra, Radare2, and performed memory forensics with Volatility/Volatility3 to identify malicious processes, injected DLLs, persistence mechanisms, and anti-analysis techniques.
- Conducted network forensics using Autopsy, FTK Imager, Wireshark, and static analysis tools to extract IOCs (IPs, domains, hashes), map C2 traffic, recover deleted artifacts, and generate threat-intel reports for SOC alert tuning.

Technical Skills

Security Domain: Cryptography, Vulnerability Management, Security Assessments, Incident Response, Endpoint Protection, Device Inventory Management, Documentation, Compliance (SOC 2, ISO 27001, GDPR, HIPAA), Data Privacy

Security Tools: Linux, Burp Suite, Nessus, Docker, WireShark, Nmap, OpenSSL/SSL, Firewalls, DNS/DHCP, NIST, SIEM, IDS/IPS, AWS, Burp suite, Cloud Security

Languages: Python, Bash, C/C++, x86/x64 Assembly, HTML/CSS, SQL

Certification: AWS fundamentals, Machine Learning Projects with Google Cloud

Research Work

An Energy Efficient System for Unmanned Aerial Vehicle-based Surveillance

- Developed an adaptive energy-efficient surveillance system for UAVs utilizing YOLO V7, with improvements through quantization methods and adaptive bit precision adjustments.